

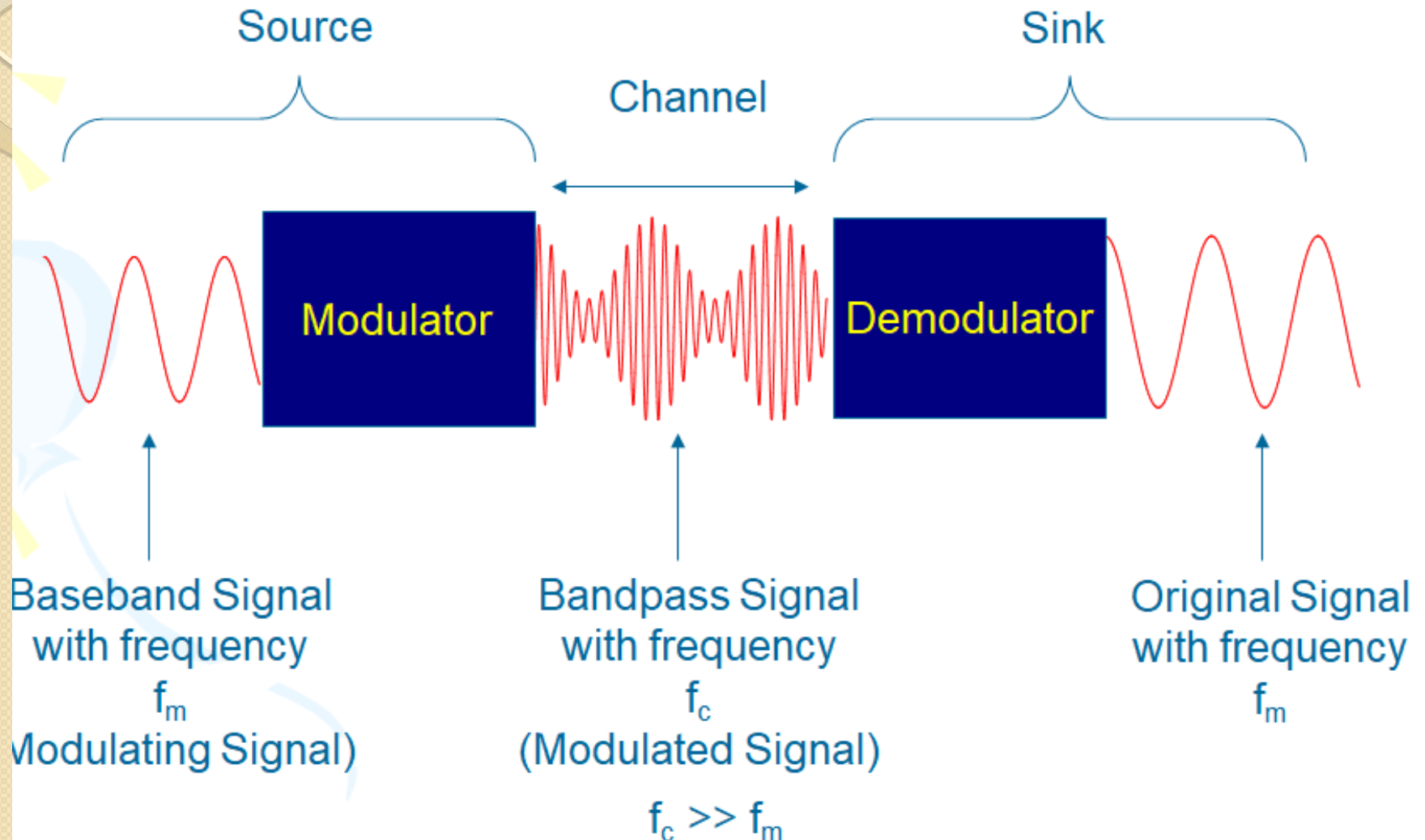


Chapter 2

Amplitude Modulation and Demodulation

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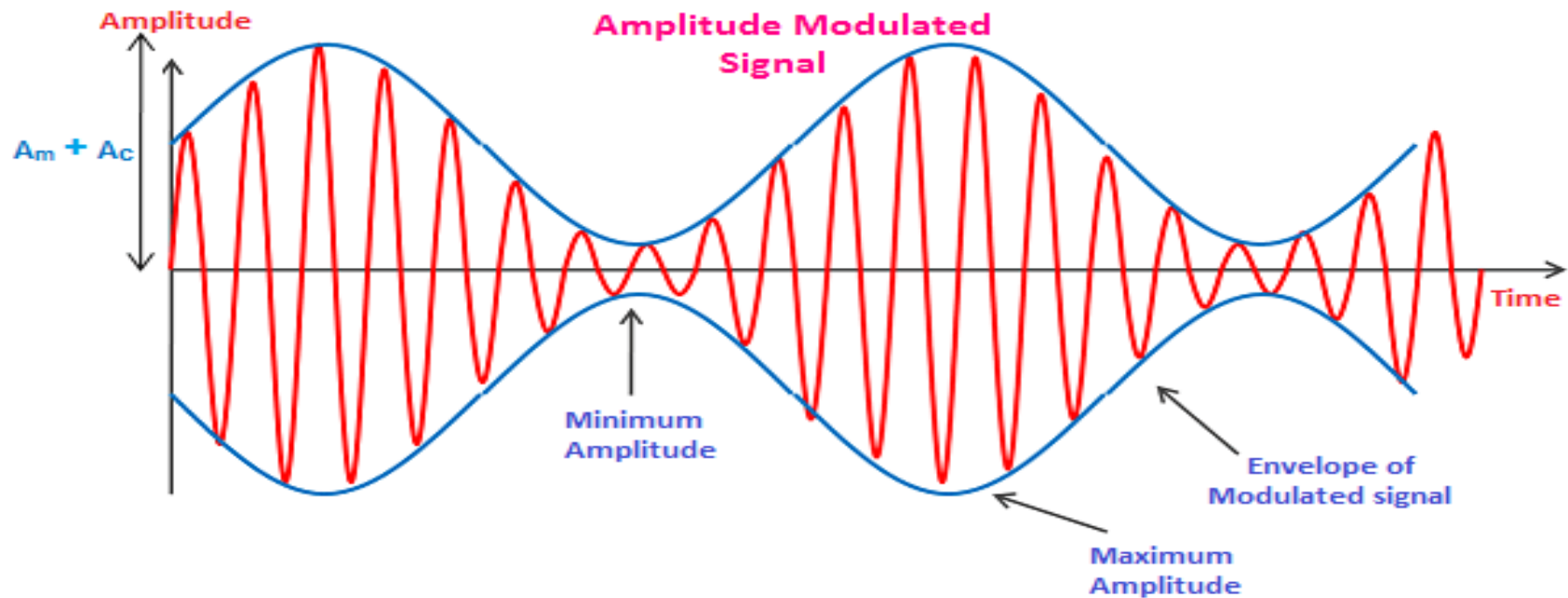
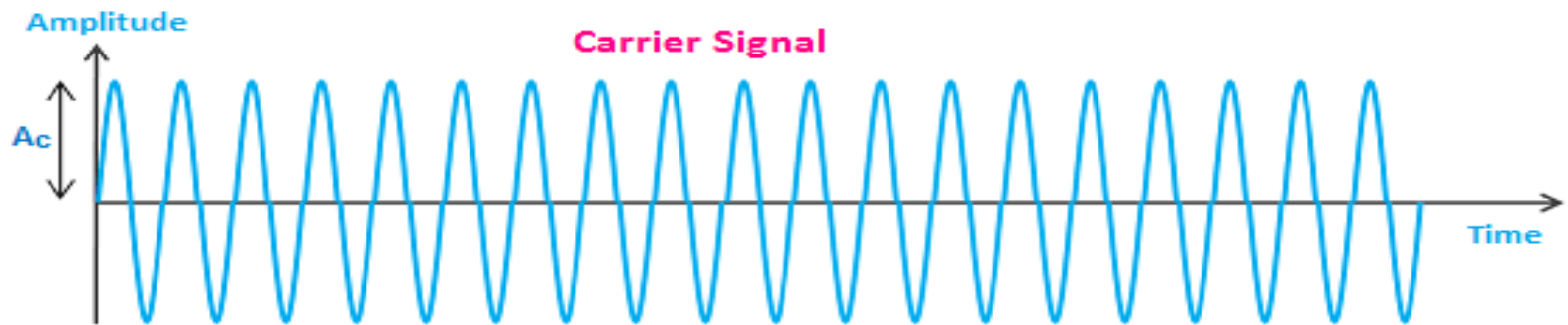
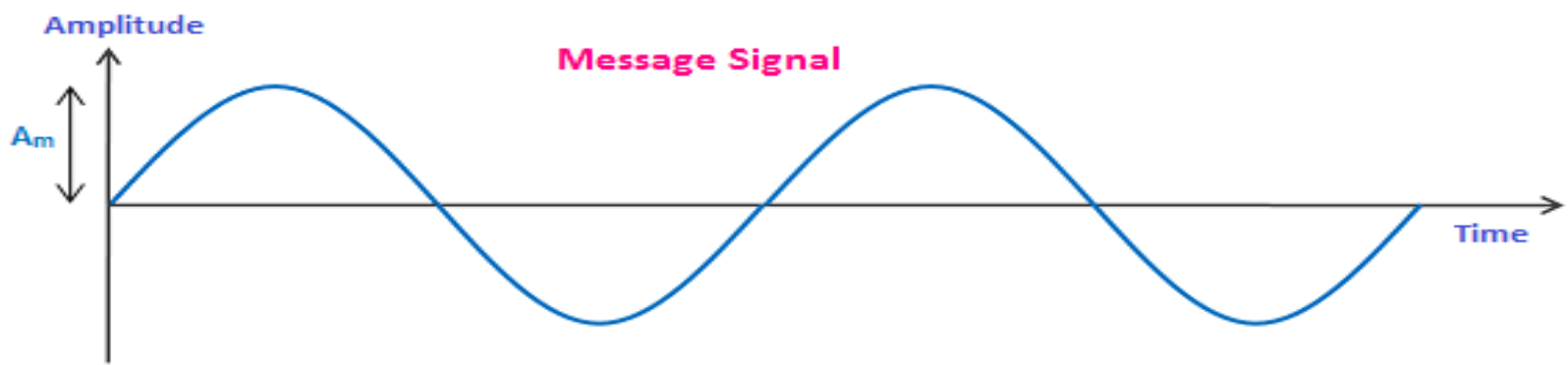
AM Modulation and Demodulation



Voice: 300-3400Hz GSM Cell phone: 900/1800MHz

Definition of Modulation

- A Modulation is the process by which **some characteristics of a carrier wave** is varied in accordance with the instantaneous value of a **modulating signal or message signal**.
- The carrier wave is changed in accordance with the **intensity (i.e amplitude) of the message signal**.
- The resultant wave is called **modulated wave** or radio wave.



Amplitude Modulation Basics

Process of changing **amplitude** of **carrier** signals in accordance with **instantaneous** values of **modulating signal(information)**, keeping **frequency** and **phase** of carrier signal **constant**

Amplitude Modulation Basics

- AM is relatively inexpensive, low quality form of modulation.
- Used for commercial broadcasting of both audio and video signals.
- Two way mobile communications such as citizens band (CB) radio.

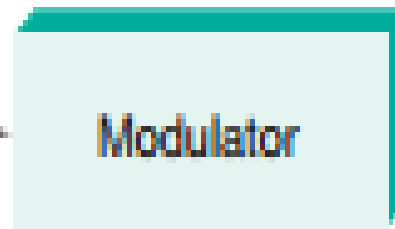
Amplitude Modulation basics

- Modulators are non linear device
- One input is a single high frequency carrier of constant amplitude
- And second input is low frequency signal.
- Voice communication utilizes range of information frequencies between 300Hz to 3000Hz.

Amplitude Modulation basics

Information
or
modulating
signal

V_m



Modulator

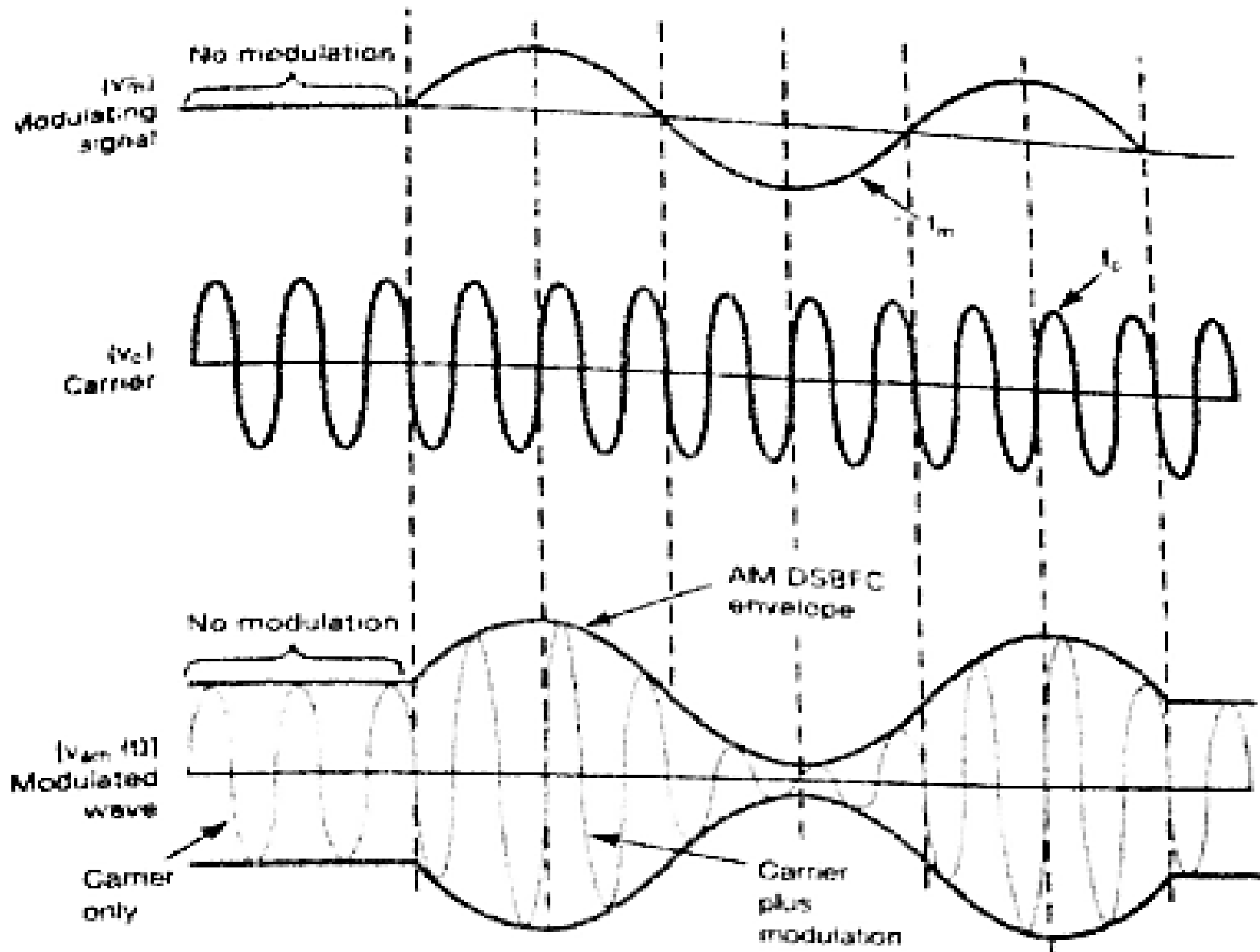
Output

$$v_2 = V_c \sin 2\pi f_c t + V_m \sin 2\pi f_m t (\sin 2\pi f_c t)$$

Carrier
signal

V_c

$$v_{AM}(t) = [V_c + V_m \sin(2\pi f_m t)][\sin(2\pi f_c t)]$$



Derivation of AM

Let modulating signal be

$$V_m = V_m \sin \omega_m t$$

$V_m \rightarrow$ Instantaneous Voltage

$V_m \rightarrow$ Peak Voltage

$\sin \rightarrow$ Nature of waveform

$\omega_m \rightarrow$ frequency $\rightarrow 2\pi f_m$

$t \rightarrow$ time

$$V_m = V_m \sin 2\pi f_m t$$

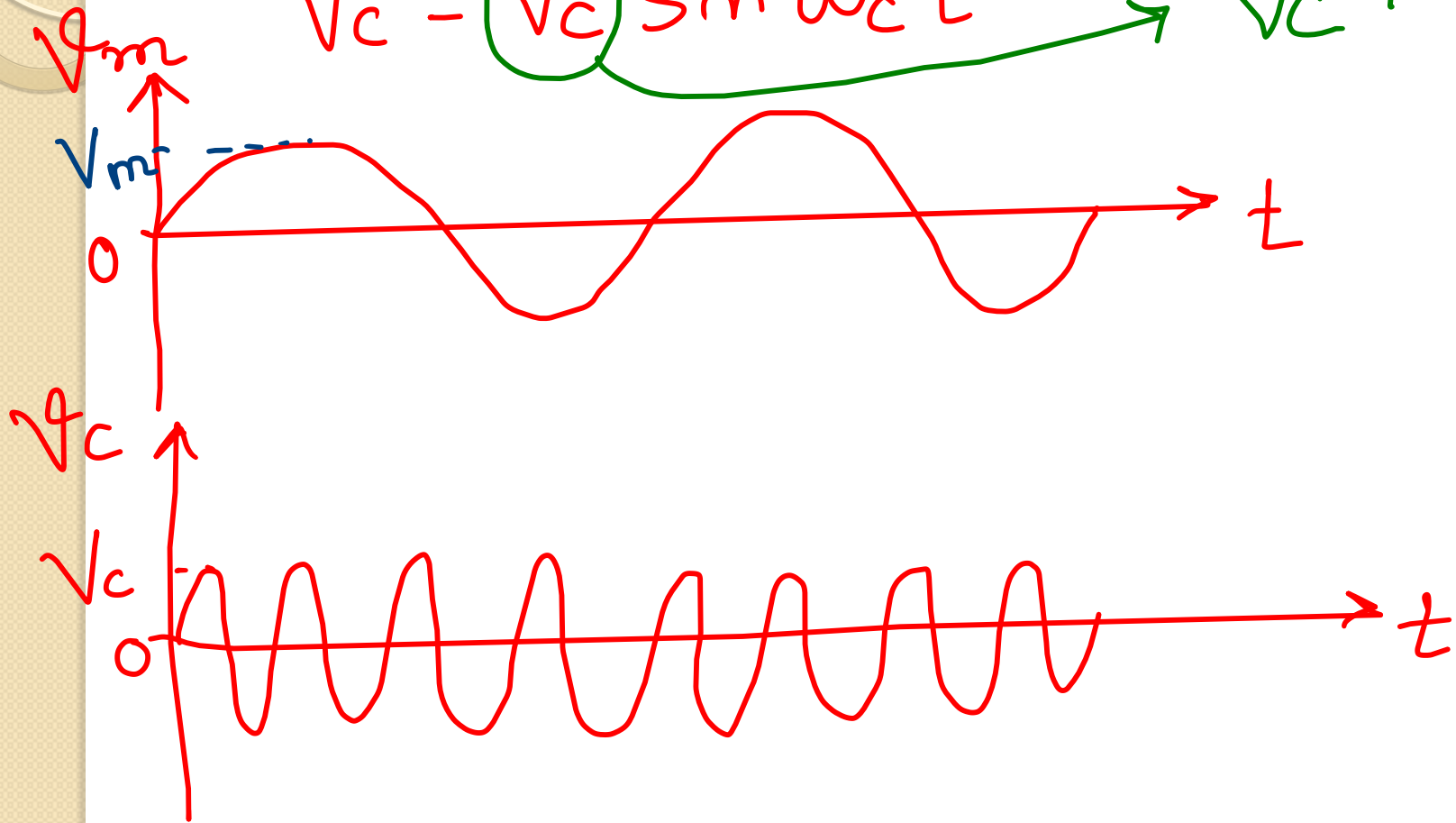
~~2 π f_m~~

Derivation of AM

Let Carrier Signal be

$$V_c = V_c \sin \omega_c t$$

$$V_c + V_m \cos \omega_m t$$



Derivation of AM

Amplitude of Am wave is

$$A = V_c + V_m$$

$$A = V_c + V_m \sin \omega_m t$$

$$A = V_c \left[1 + \frac{V_m}{V_c} \sin \omega_m t \right]$$

$$A = V_c [1 + m \sin \omega_m t] \text{ — (1)}$$

where, $m = \frac{V_m}{V_c} \rightarrow$ Modulation index /
Depth of modulation

Derivation of AM

Instantaneous amplitude of AM signal is

$$V_{am}(t) = A \sin \omega_c t$$

$$V_{am}(t) = V_c [1 + m \sin \omega_m t] \sin \omega_c t$$

$$V_{am}(t) = V_c \sin \omega_c t + m V_c \underbrace{\sin \omega_c t}_{\sin A} \cdot \underbrace{\sin \omega_m t}_{\sin B}$$

Note:

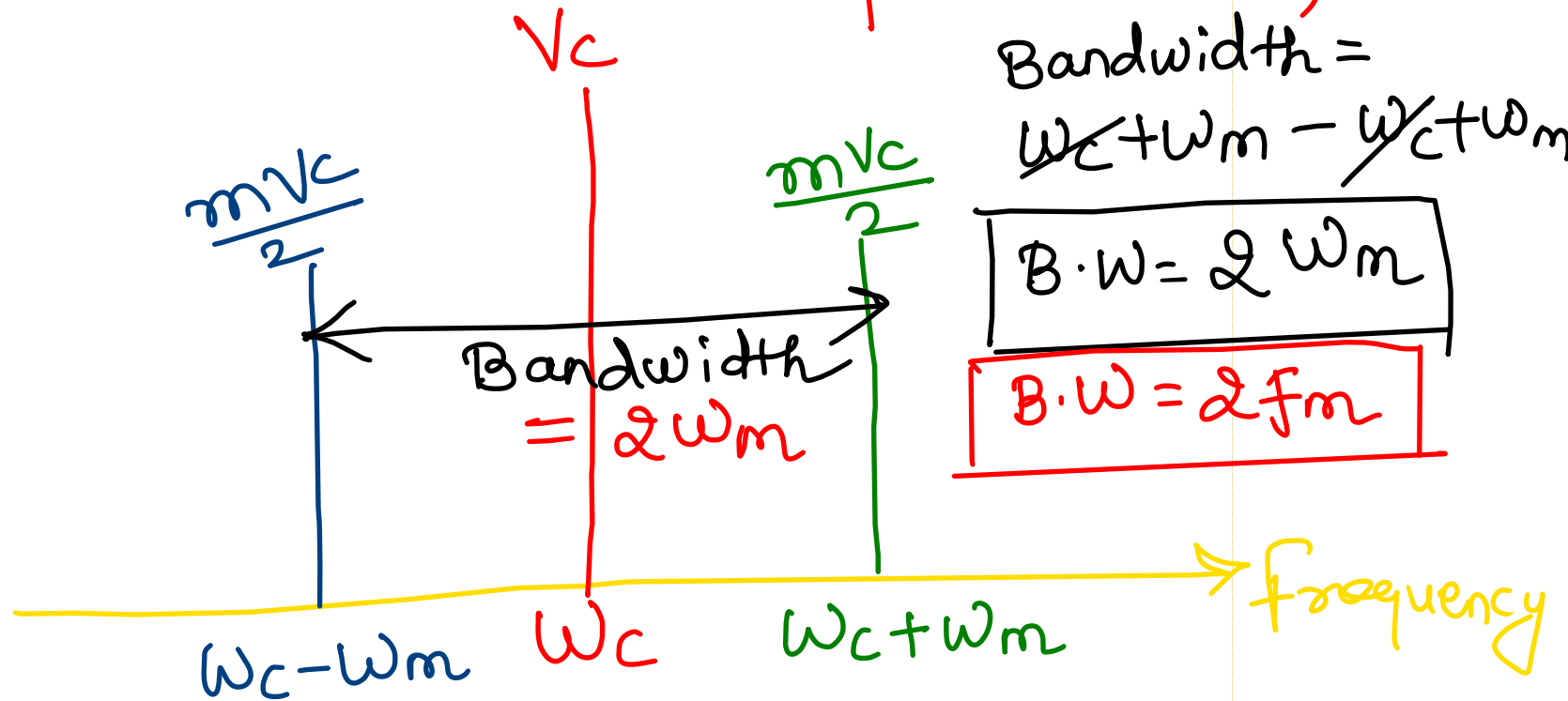
$$A = \omega_c t, B = \omega_m t$$

$$\sin A \cdot \sin B = \frac{1}{2} [\cos(A-B) - \cos(A+B)]$$

Derivation of AM

$$V_{am}(t) = \underbrace{V_c \sin \omega_c t}_{\text{Carrier Signal}} + \underbrace{\frac{mV_c}{2} \cos(\omega_c - \omega_m) t}_{\text{Lower Side band (LSB)}} - \underbrace{\frac{mV_c}{2} \cos(\omega_c + \omega_m) t}_{\text{Upper Sideband (USB)}}$$

Derivation of AM (Frequency Domain representation)



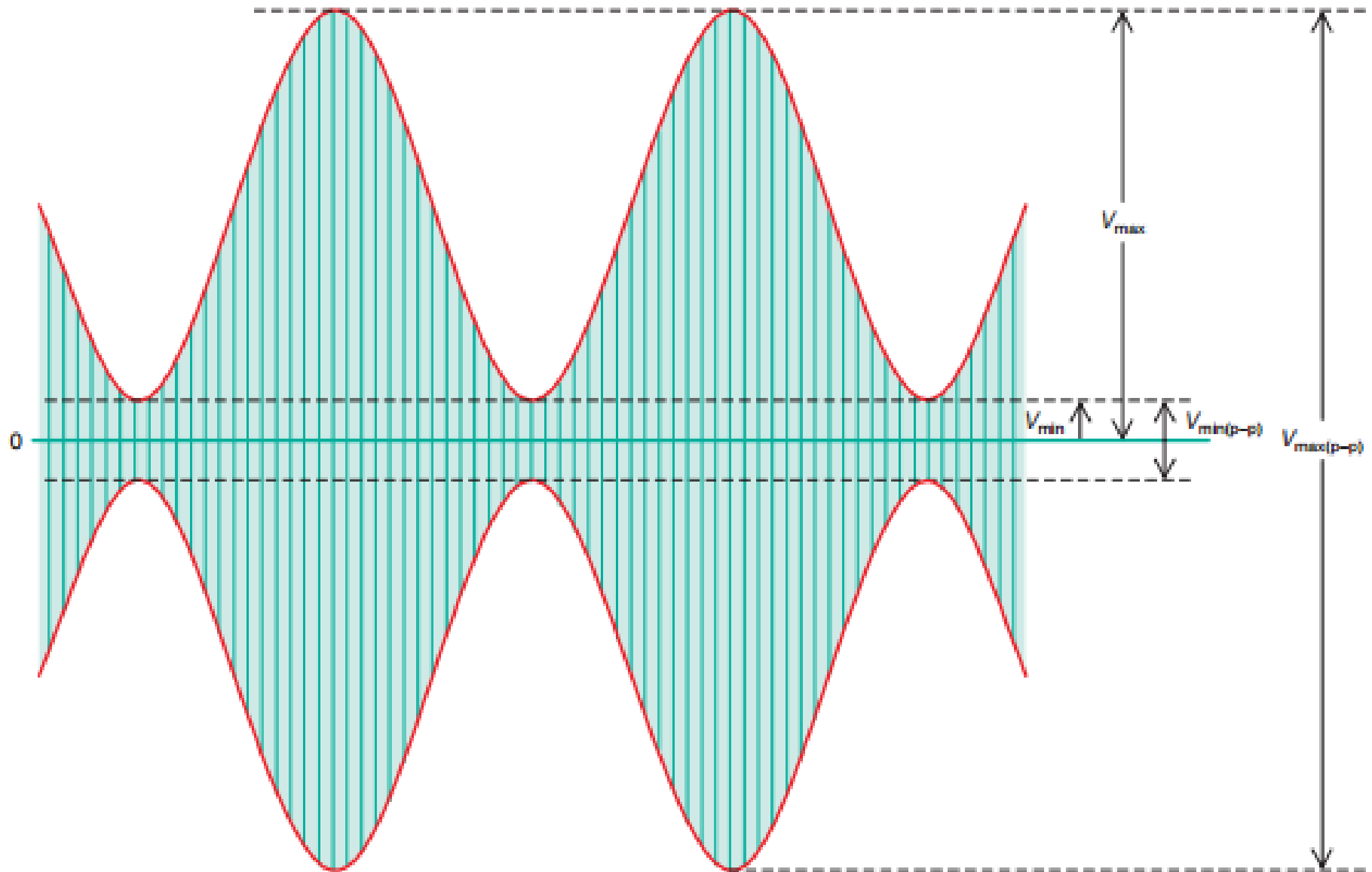
Modulation Index

- It is Ratio of modulating signal voltage (V_m) to carrier signal voltage (V_c).

$$m = \frac{V_m}{V_c}$$

- Value of m will be between 0 and 1
- $m < 1$ is called under modulation
- $m = 1$ is called 100% modulation
- $m > 1$ is called over modulation

Modulation Index



Modulation Index

$$V_m = \frac{V_{\max} - V_{\min}}{2}$$

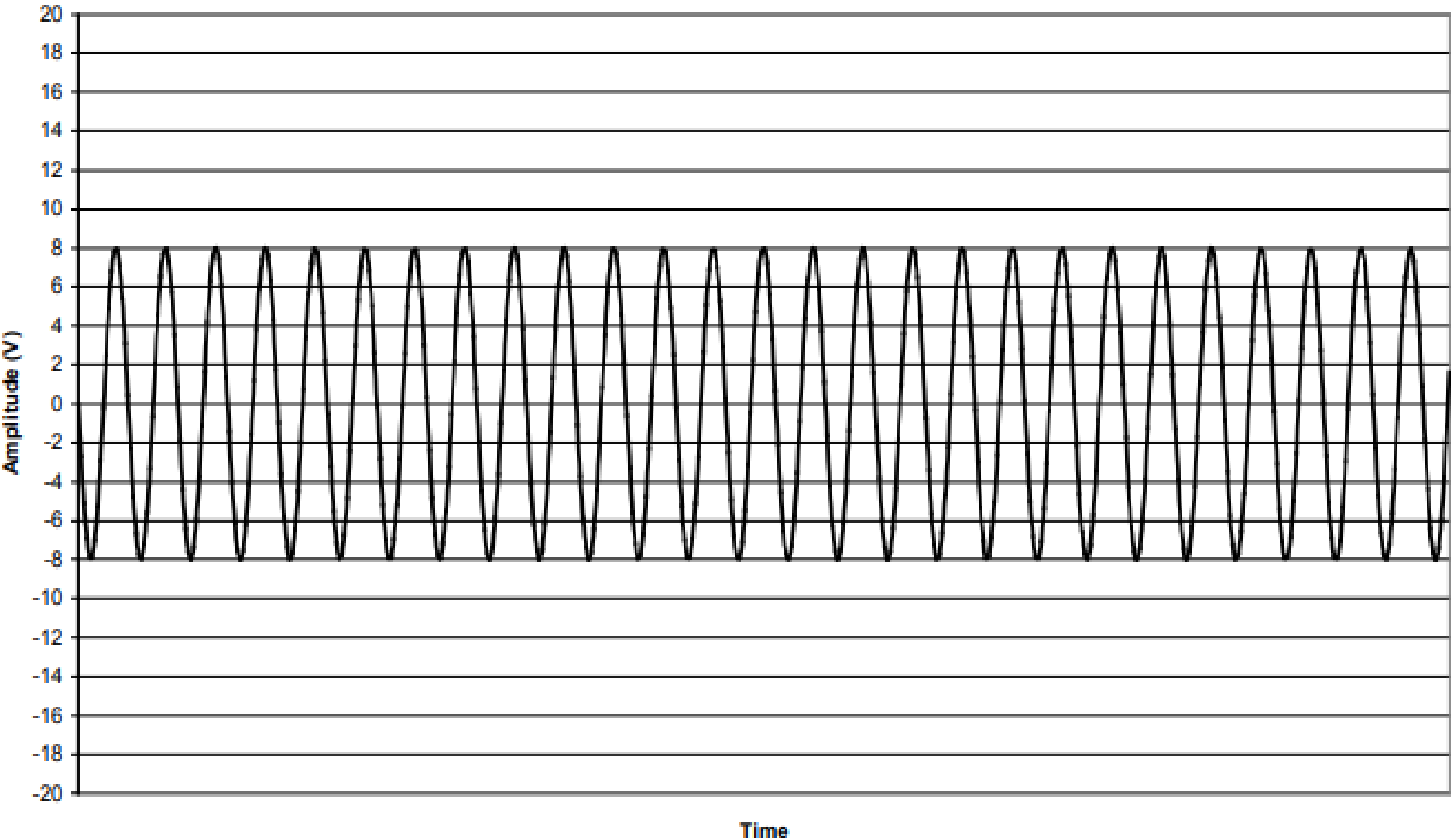
$$V_c = \frac{V_{\max} + V_{\min}}{2}$$

$$m = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}}$$

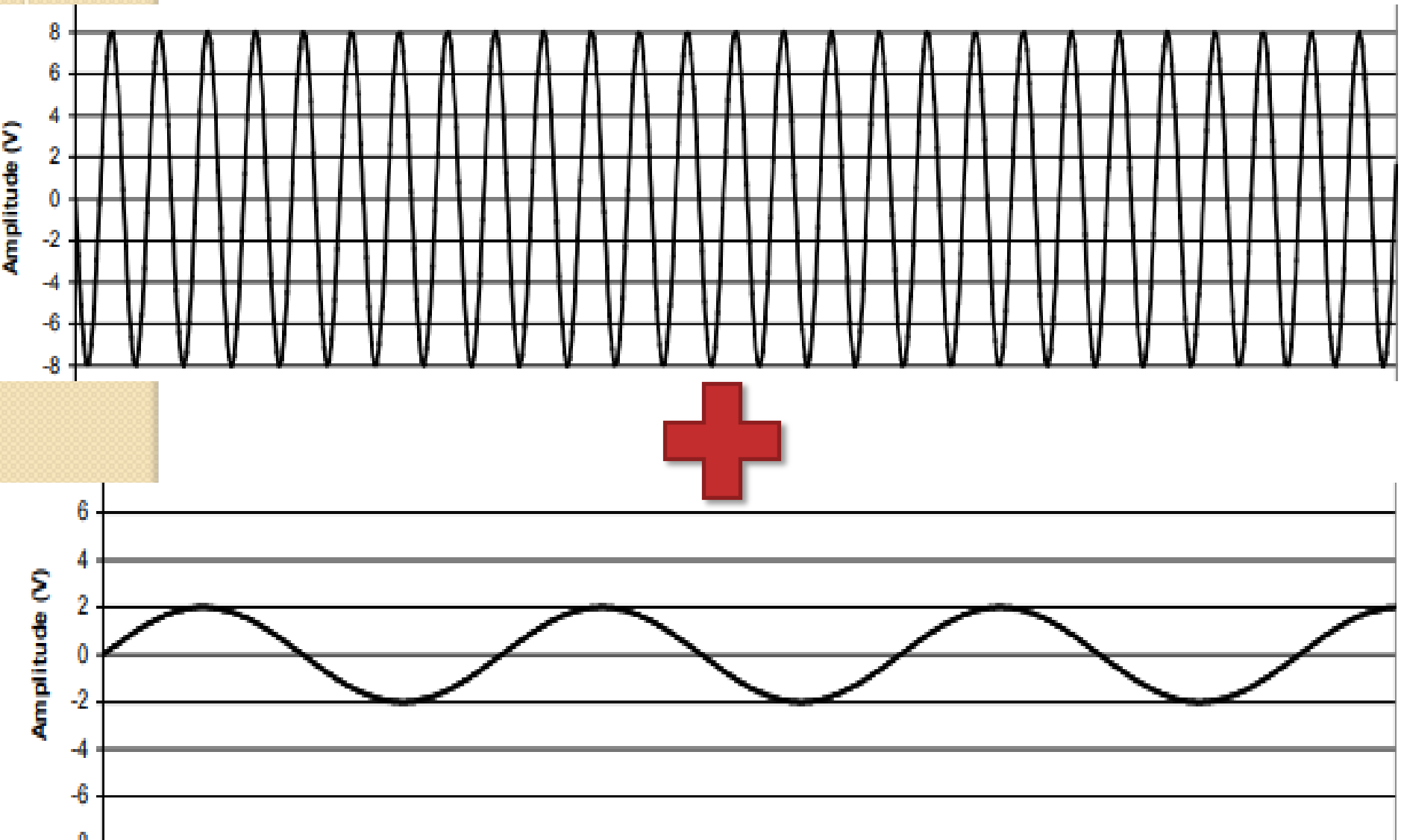


Depth of Modulation ($V_c=8V$)

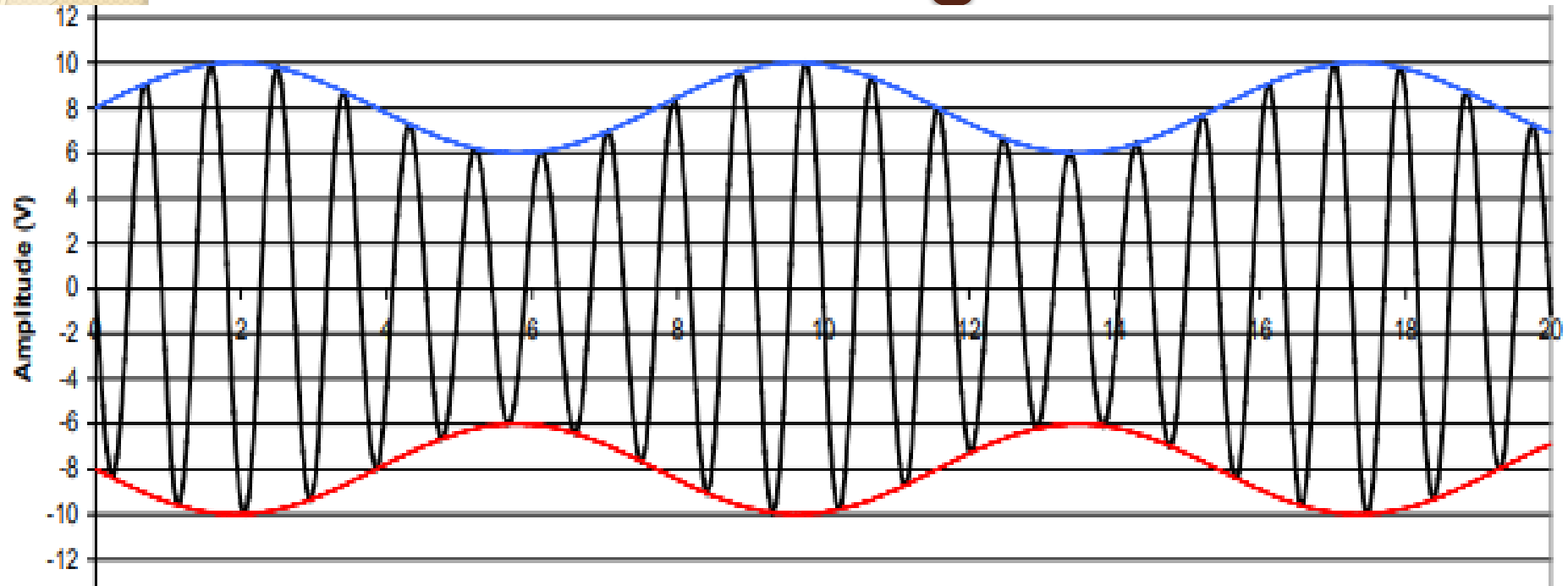
Carrier Wave



Case I: Add information signal of 2V to carrier



Resultant Signal

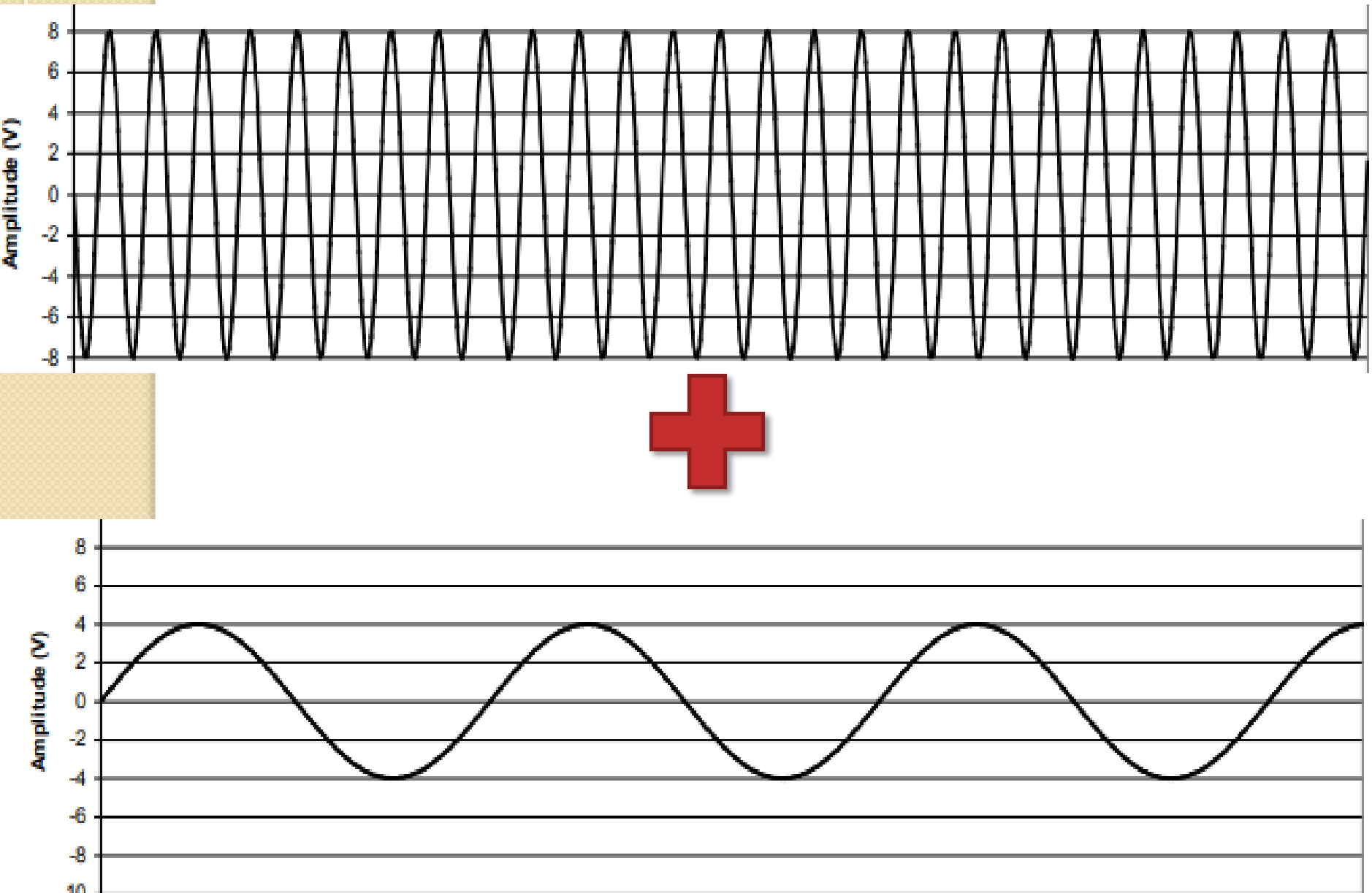


$$V_{\max} = V_c + V_m = 8 + 2 = 10 \text{ V}$$

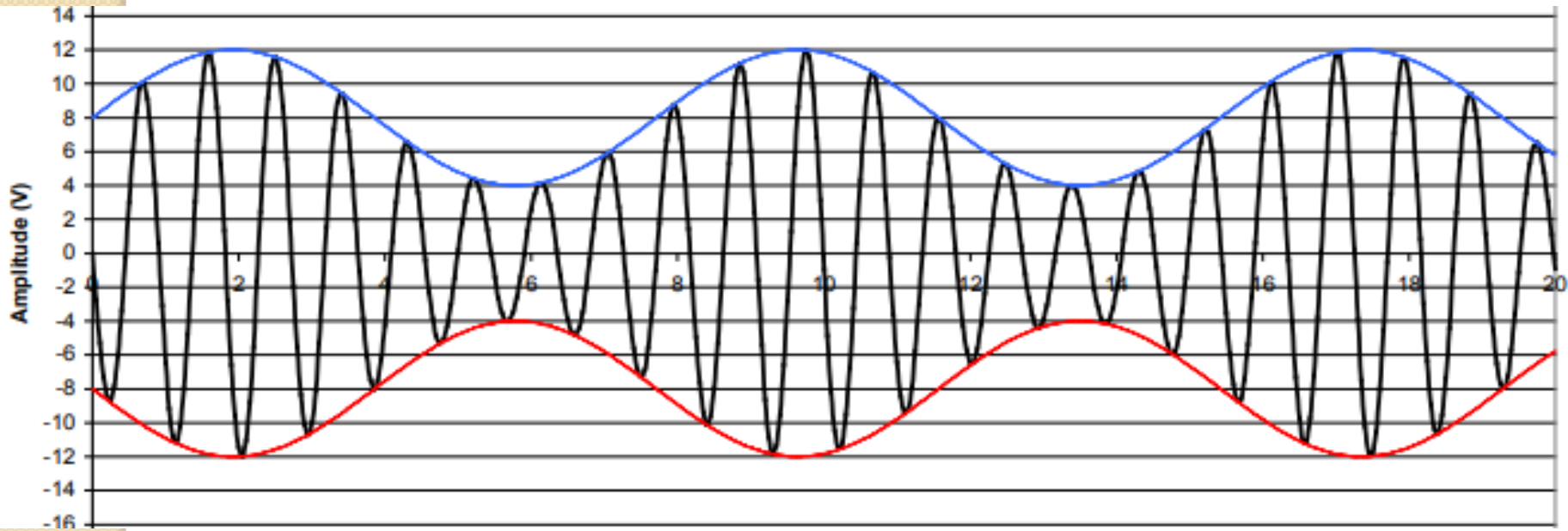
$$V_{\min} = V_c - V_m = 8 - 2 = 6 \text{ V}$$

$$m = \frac{4}{16} \times 100\% = 25\%$$

Case 2: Add information signal of 4V to carrier



Resultant Signal

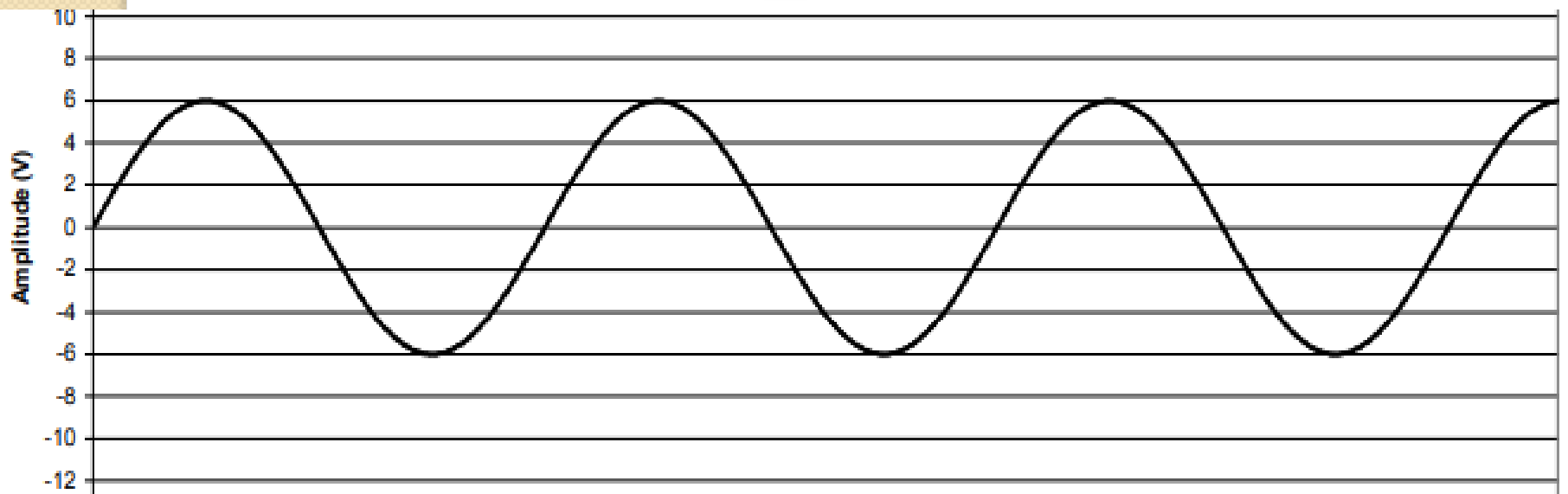
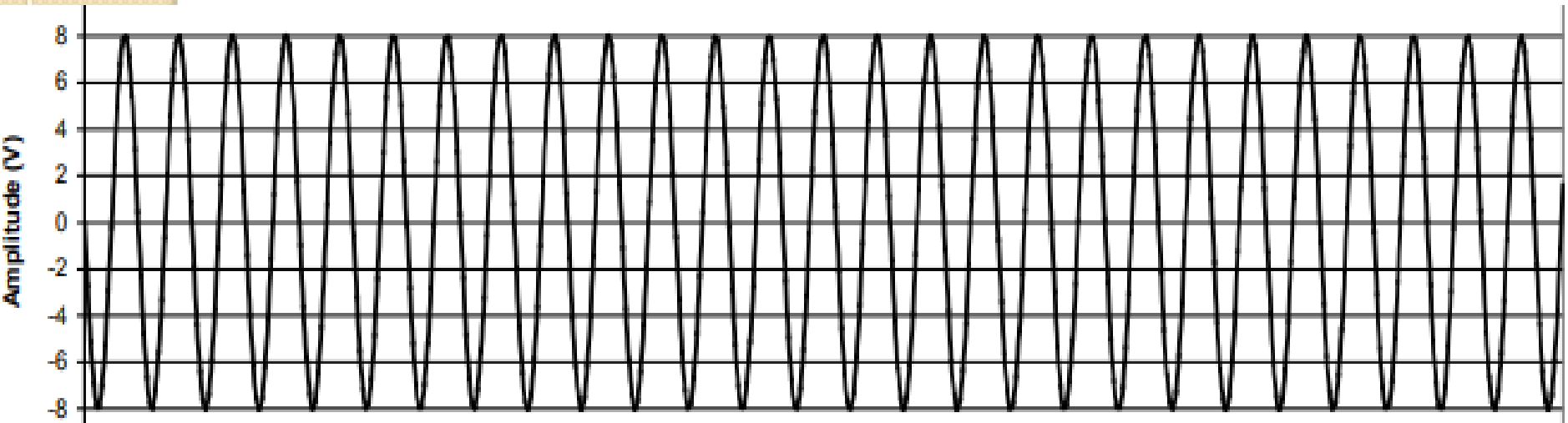


$$V_{\max} = V_c + V_m = 8 + 4 = 12 \text{ V}$$

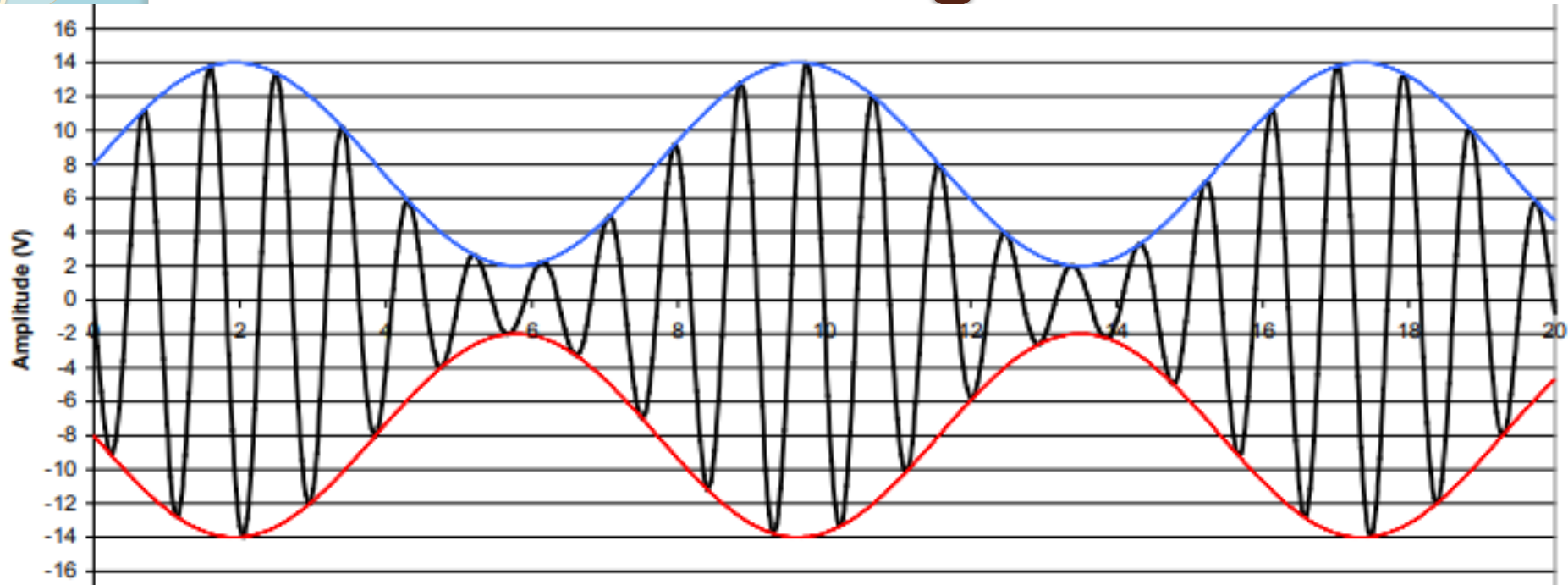
$$V_{\min} = V_c - V_m = 8 - 4 = 4 \text{ V}$$

$$m = \frac{8}{16} \times 100\% = 50\%$$

Case 3: Add information signal of 6V to carrier



Resultant Signal

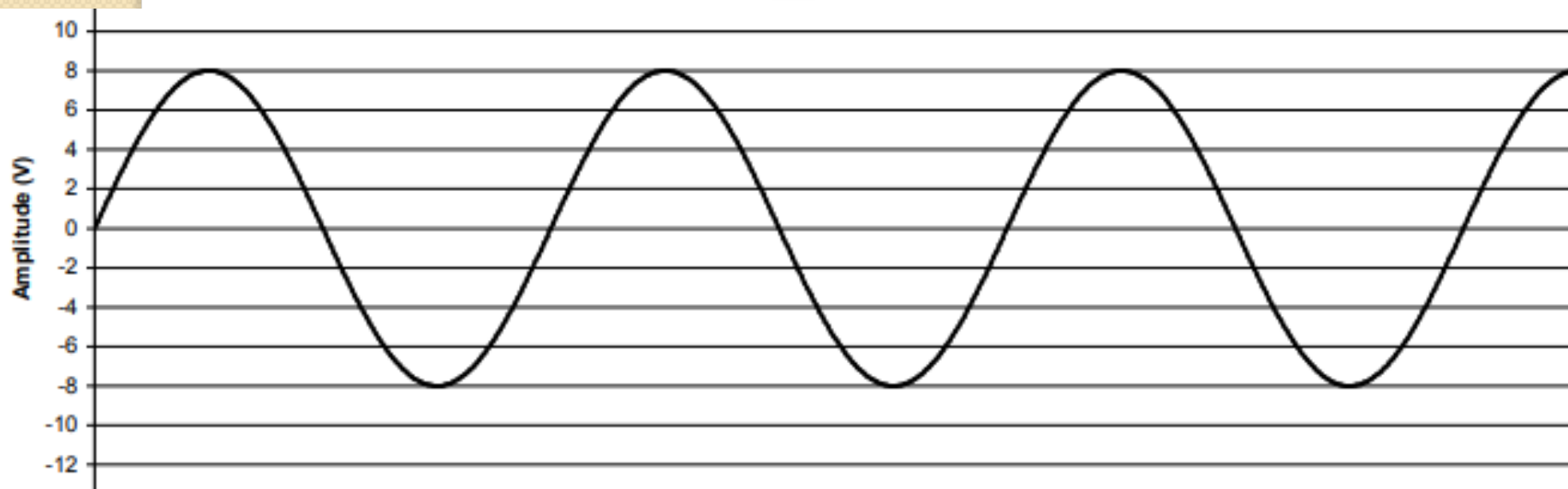
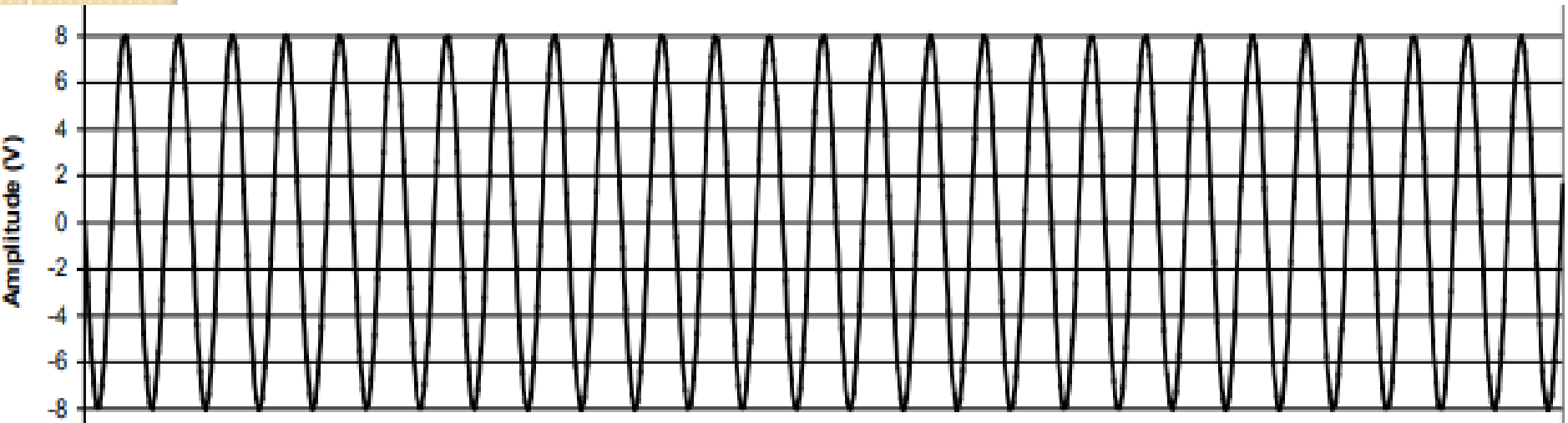


$$V_{\max} = V_c + V_m = 8 + 6 = 14 \text{ V}$$

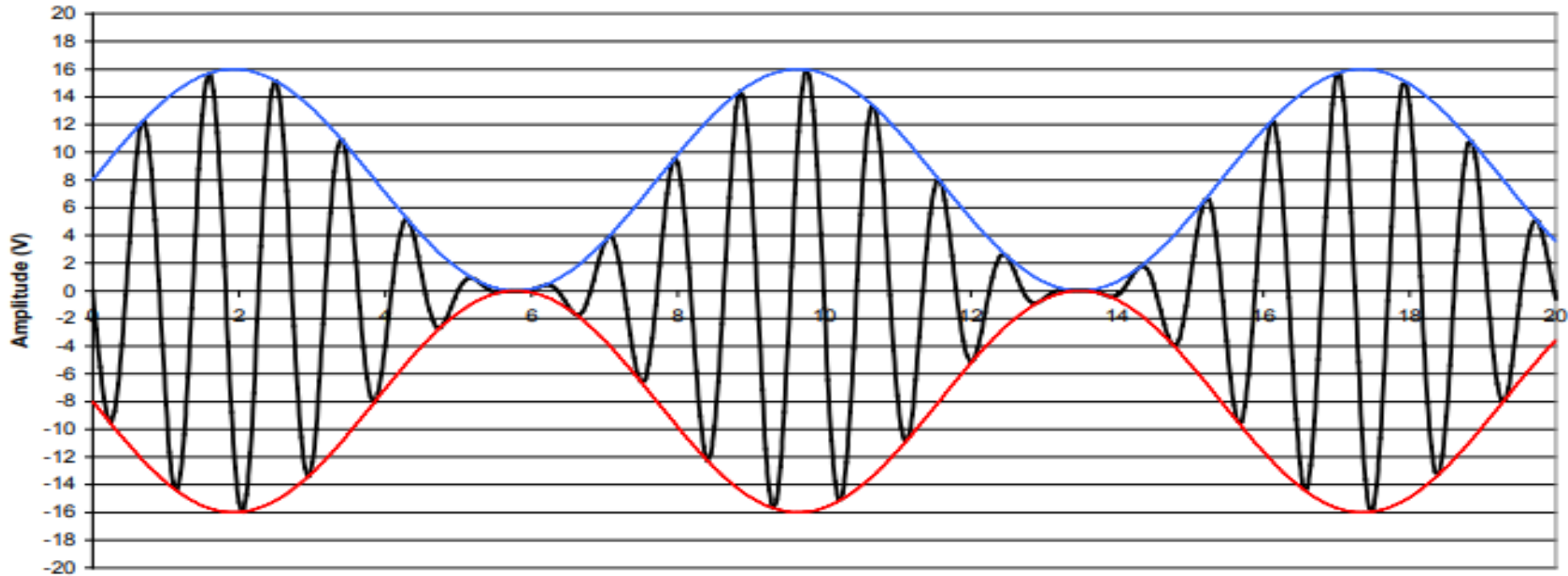
$$V_{\min} = V_c - V_m = 8 - 6 = 2 \text{ V}$$

$$m = \frac{12}{16} \times 100\% = 75\%$$

Case 4: Add information signal of 8V to carrier



Resultant Signal

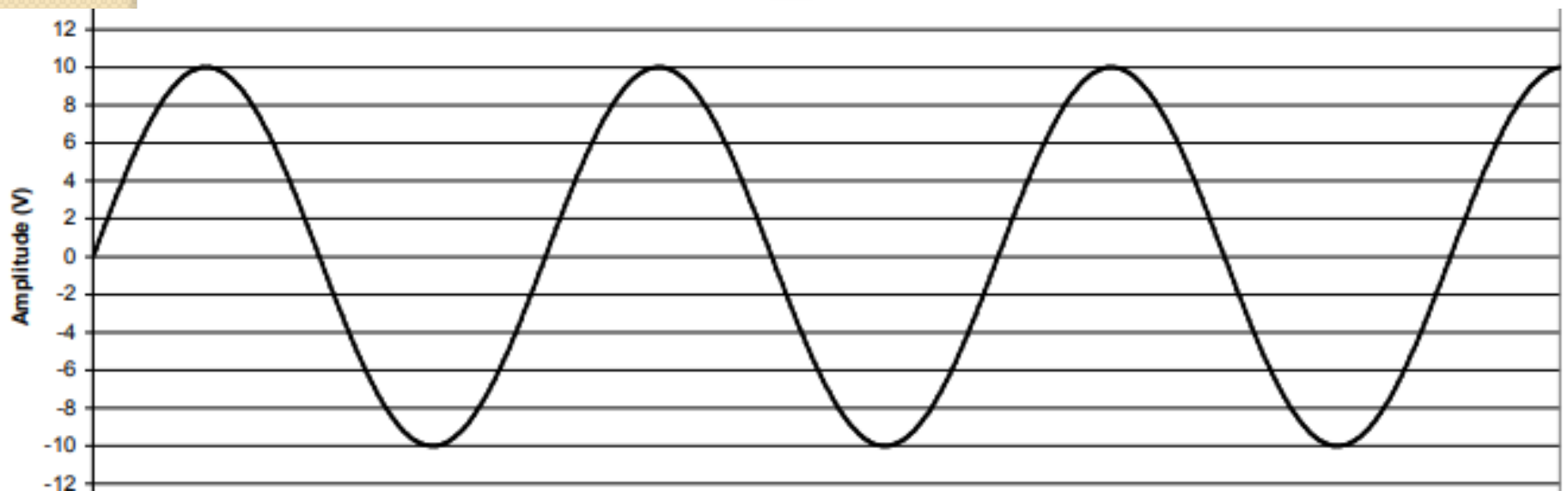
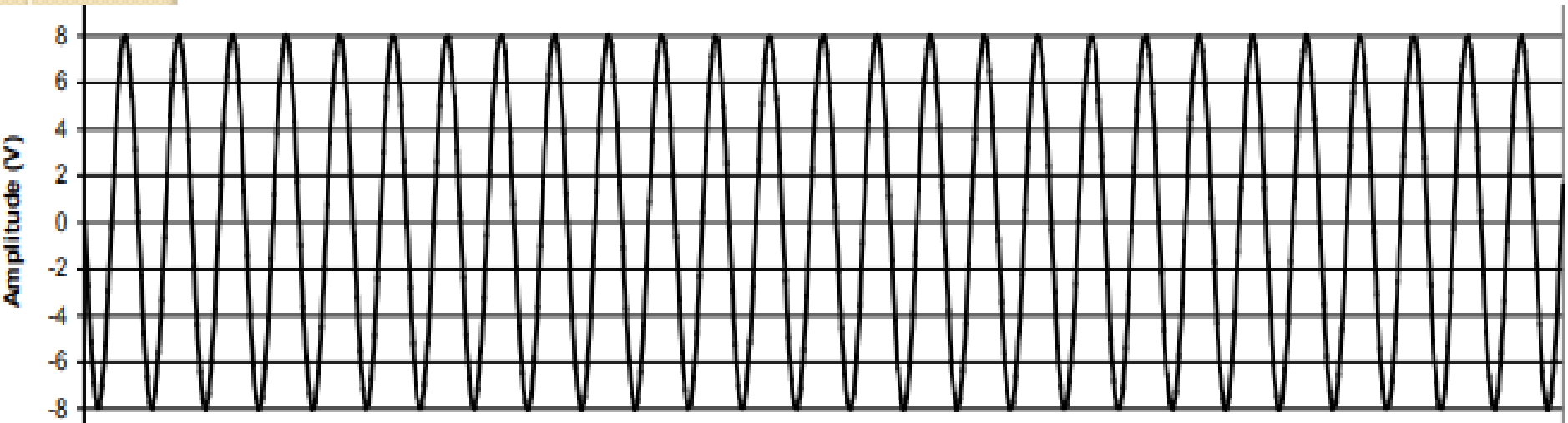


$$V_{\max} = V_c + V_m = 8 + 8 = 16 \text{ V}$$

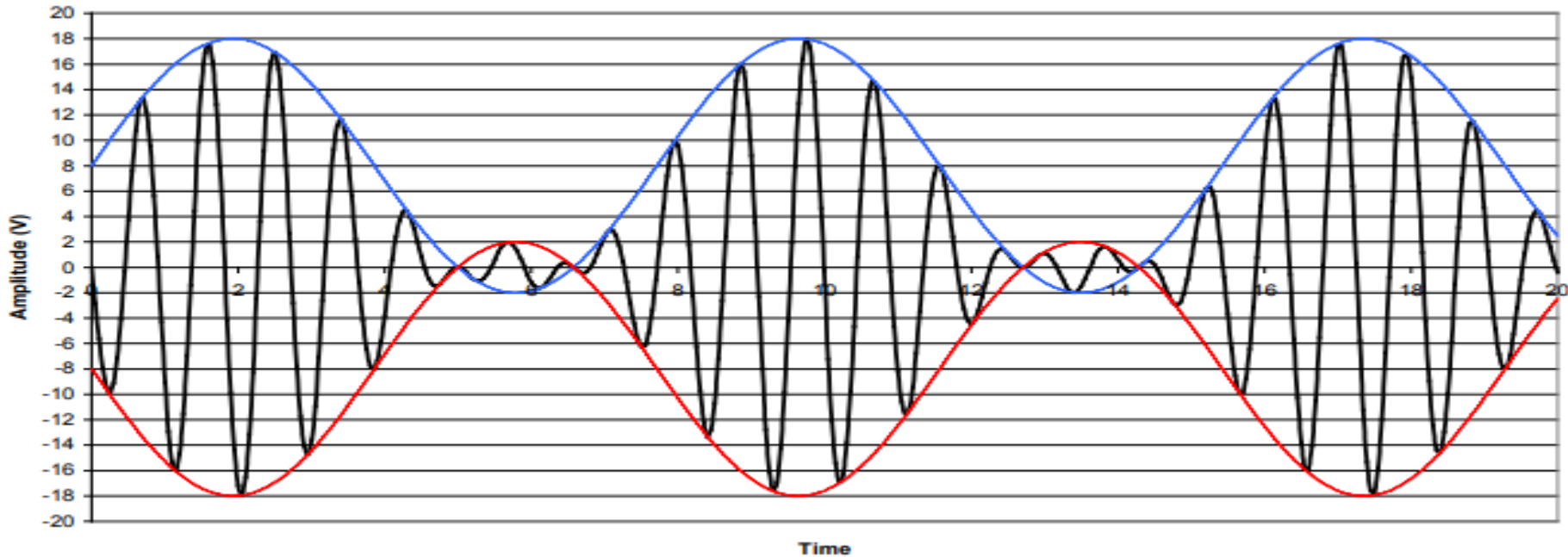
$$V_{\min} = V_c - V_m = 8 - 8 = 0 \text{ V}$$

$$m = \frac{16}{16} \times 100\% = 100\%$$

Case 5: Add information signal of 10V to carrier



Resultant Signal

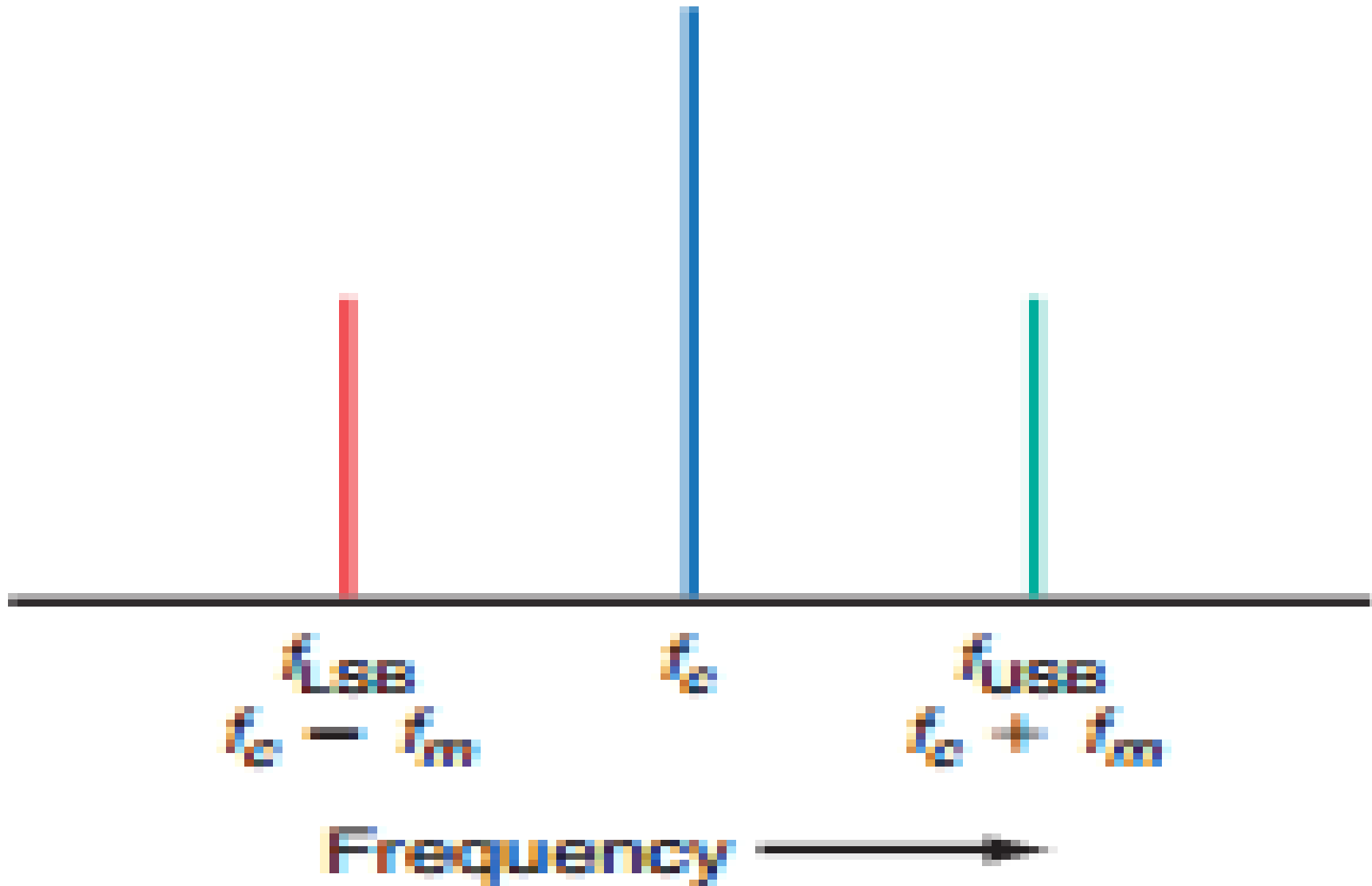


$$V_{\max} = V_c + V_m = 8 + 10 = 18 \text{ V}$$

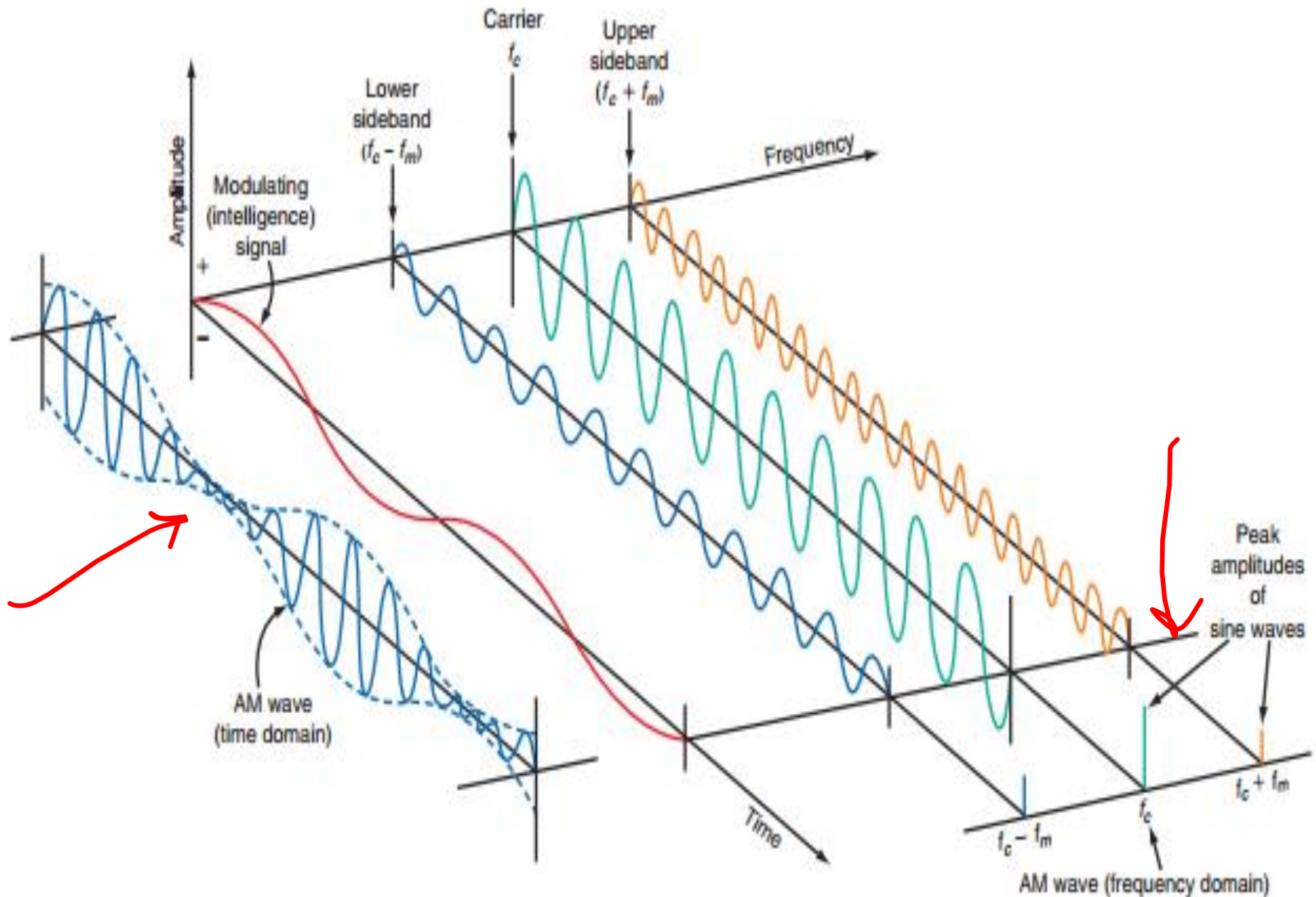
$$V_{\min} = V_c - V_m = 8 - 10 = -2 \text{ V}$$

$$m = \frac{20}{16} \times 100\% = 125\%$$

Frequency Spectrum of AM



Frequency Spectrum of AM



AM Derivation Practice

Q1. Assume $m(t) = A_m \sin \omega_m t$ and $c(t) = A_c \sin \omega_c t$

Derive expression for amplitude modulated (AM) wave AM(t).

Answer: $A_c \sin \omega_c t + \frac{A_c m_a}{2} \cos 2\pi(f_c - f_m)t - \frac{A_c m_a}{2} \cos 2\pi(f_c + f_m)t$

Q2. Assume $m(t) = A_m \sin \omega_m t$ and $c(t) = A_c \cos \omega_c t$

Derive expression for amplitude modulated (AM) wave AM(t).

Answer: $A_c \cos \omega_c t + \frac{A_c m_a}{2} \sin 2\pi(f_c - f_m)t - \frac{A_c m_a}{2} \sin 2\pi(f_c + f_m)t$

Q3. Assume $m(t) = A_m \cos \omega_m t$ and $c(t) = A_c \sin \omega_c t$

Derive expression for amplitude modulated (AM) wave AM(t).

Answer: $A_c \sin \omega_c t + \frac{A_c m_a}{2} \sin 2\pi(f_c + f_m)t + \frac{A_c m_a}{2} \sin 2\pi(f_c - f_m)t$

Power Calculations of Am

$$P_t = P_c + P_{USB} + P_{LSB}$$

$$\begin{aligned} P &= V \cdot I \\ &= V \cdot \frac{V}{R} \\ P &= \frac{V^2}{R} \end{aligned}$$

$$P = \frac{V_{rms}^2}{R} = \frac{(V_m / \sqrt{2})^2}{R}$$

$V_{rms} \rightarrow$ RMS value of voltage

$V_m \rightarrow$ Peak value of voltage

Carrier Power $P_C = \frac{(V_C/\sqrt{2})^2}{R} = \frac{V_C^2}{2R} \rightarrow \text{Component 1}$

Upper Sideband Power $P_{USB} = \frac{\left(\frac{mV_C}{2}/\sqrt{2}\right)^2}{R} = \frac{m^2 V_C^2}{8R} \rightarrow \text{Component 2}$

Lower Sideband Power $P_{LSB} = \frac{\left(\frac{mV_C}{2}/\sqrt{2}\right)^2}{R} = \frac{m^2 V_C^2}{8R} \rightarrow \text{Component 3}$

$$P_t = P_c + P_{USB} + P_{LSB}$$

$$= \frac{V_c^2}{2R} + \frac{m^2 V_c^2}{8R} + \frac{m^2 V_c^2}{8R}$$

$$P_t = \frac{V_c^2}{2R} \left(1 + \frac{m^2}{4} + \frac{m^2}{4} \right)$$

$$P_t = \frac{V_c^2}{2R} \left(1 + \frac{m^2}{2} \right)$$

$$P_t = P_c \left(1 + \frac{m^2}{2} \right)$$

Case-1

$$m = 0 \quad (\text{No modulation})$$

$$P_t = P_c$$

Case-2

$$m = 1 \quad (100\% \text{ modulation})$$

$$P_t = 1.5 P_c$$